Description

Illustration





Figure caption

1. Decorative terracotta floor tiles. 2. Roof of traditional Spanish house covered in terracotta tiles. Images used under license from Shutterstock.com

The material

The name "terracotta" comes from the Italian for "baked earth". It is a semifired waterproof ceramic clay, used to make bricks and shingles.

Compositional summary

Mixed silicates.

General properties

Density		1.9e3	-	2.75e3	kg/m^3
Price	*	0.43	-	0.642	GBP/kg
Date first used		-10000			

Material form that data applies to

Bulk	✓

Building system

Superstructure	✗
Enclosure	✓
Interiors	✓
Services	✗

Mechanical properties

Young's modulus	*	4	-	8	GPa

Shear modulus	*	2	-	4	GPa
Bulk modulus	*	3	-	6	GPa
Bending modulus	*	5	-	9	GPa
Poisson's ratio	*	0.2	-	0.25	
Yield strength (elastic limit)	*	15	-	20	MPa
Tensile strength	*	15	-	20	MPa
Compressive strength		25	-	28	MPa
Bending strength		16.6	-	29.5	MPa
Elongation	*	0.19	-	0.5	% strain
Hardness - Vickers		11	-	13	HV
Fatigue strength at 10^7 cycles	*	6	-	8	MPa
Fracture toughness	*	1	-	2	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.003	-	0.008	

Thermal and combustion properties

Thermal conductor or insulator?	Poor insulator			
Thermal resistivity	0.66	-	2	m.°C/W
Thermal expansion coefficient	5	-	10	µstrain/°C
Specific heat capacity	760	-	820	J/kg.°C
Maximum service temperature	673	-	777	°C
Flammability	Non-flammable			
Emissivity	0.6	-	0.85	

Hygro-thermal properties

Water absorption	3	-	15	%
Frost resistance	Poor			

Electrical properties

Electrical conductor or insulator?	Good insulator				
Electrical resistivity	*	1e14	-	3e14	µohm.cm

Optical properties

Natural color

Brown to orange

Transparency	Opaque			
Transmissivity	0			%

Critical Materials Risk

High critical material risk?

	Yes
Notes	Al (aluminum) added to the 2018 US critical minerals list

Acoustic properties

Sound absorption	Average
Sound isolation	Good

Processability (scale 1 = impractical to 5 = excellent)

Moldability	2	-	4	
Machinability	1	-	2	

Durability

Water (fresh)	Acceptable
Water (salt)	Excellent
Weak acids	Excellent
Strong acids	Excellent
Weak alkalis	Acceptable
Strong alkalis	Unacceptable
Organic solvents	Excellent
UV radiation (sunlight)	Excellent
Wear resistance	Excellent
Industrial atmosphere	Acceptable
Rural atmosphere	Excellent
Marine atmosphere	Excellent

Primary material production: energy and climate change

Climate change (CO_2-eq) (virgin grade production)	* 1.82	-	2.01	kg/kg
Embodied energy (virgin grade production)	* 13	-	14.4	MJ/kg
Water Usage	* 5.27	-	5.83	l/kg

Material processing: energy

Machining, grinding, embodied energy	* 18.4	-	19.4	MJ/kg

Material processing: climate change

Machining, grinding, climate change (CO_2-eq)	* 1.21	-	1.27	kg/kg

Recycling and end of life

Recycle	✓				
Functional recycle	✗				
Climate change (CO_2-eq), recycling	* 0.182	-	0.201	kg/kg

Embodied energy, recycling	* 1.82	-	2.02	MJ/kg
Recycle fraction in current supply	0			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
A renewable resource?	✗			

Supporting information

Typical uses

as tiles - for roofing and cladding. terracotta can also be formed into more complex shapes to form sculptures and pottery pieces.

Tradenames

Terracade

Links

ProcessUniverse

Reference